

Strategic Risk Management (SRM) Benchmarking Q&A

Expert Consultation Questionnaire

Benchmarking and Modular Integration of Predictive SRM in EV Battery Supply Chains

Confidentiality: All responses are anonymized and optional. Respondents are referenced by professional role category only. No names, employing organizations, or contact details are recorded in any published output. GDPR-compliant.

Estimated Completion Time: 30–40 minutes

Instructions: Please respond based on your professional experience and knowledge. There are no correct or incorrect answers — your expert judgement is the primary input sought. For scaled items, please use the provided rating scale and add a brief comment where indicated.

SECTION A — Respondent Profile

For sampling documentation and purposive sampling verification. Not linked to individual responses.

1. Which of the below field best describes your professional experience / expertise

- ☐ Supply Chain
- ☐ Supply Chain Risk Management
- ☐ Regulatory Compliance
- ☐ Enterprise Risk Management
- ☐ Strategy (Procurement / Business)
- ☐ Sustainability
- ☐ Critical mineral (Trading, procurement, analyst, etc.)
- ☐ Supply chain Academic / researcher
- ☐ Other

2. How many years of experience do you have in EV battery supply chains, critical mineral sourcing, or supply chain risk management ?

- ☐ 0-2
- ☐ 2-4
- ☐ 4-6
- ☐ 6-10
- ☐ 10+

3. In which supply chain tier(s) do you have professional experience/s? (Select all that apply)

- ☐ Tier-1 (Cell manufacturers)
- ☐ Tier 2 (Battery active material / components)
- ☐ Tier-3 or beyond (Raw Minerals / Material)
- ☐ OEM / Automotive

4. What is your primary region of expertise? (multiple choice)

- ☐ China
- ☐ United States / North America
- ☐ Korea / Japan
- ☐ South America
- ☐ Central Africa
- ☐ Europe / EU
- ☐ Other

SECTION B — Research Gap Validation

Purpose: Confirm whether the proposed research problem reflects a real, current industry need — before the research design is finalised

5. The proposed research addresses this core problem: *"No replicable, cross-tier instrument exists for benchmarking strategic risk management (SRM) maturity across Tier 1 cell manufacturers and Tier 3 raw material suppliers — and no evidence-grounded, modular roadmap exists for integrating predictive risk tools at different maturity levels in this sector."*

How accurately does this describe a **real gap** you observe in your professional context?

	Not a real gap	Minor gap	Significant gap	Critical gap
No cross-tier SRM benchmarking instrument exists	☐	☐	☐	☐
No modular, practical roadmap for predictive tool adoption	☐	☐	☐	☐
AI/ML tools not applied to battery-specific supply risk	☐	☐	☐	☐
ESG maturity data at Tier 3 is scarce and incomparable	☐	☐	☐	☐

6. In case the answer for above question is **Not a Real Gap / Minor Gap**, please provide a comment or provide the study / instruments that addresses

7. To your knowledge, does any tool, standard, or published study **already address** cross-tier SRM benchmarking for battery supply chains in a replicable, comparable way?

- ☐ No — nothing comparable exists
- ☐ Partially — something similar exists: (Please describe briefly)
- ☐ Yes — the following addresses it fully: (Please describe briefly)

8. In case the answer for above question is **Partially / Yes**, please provide provide the study / instruments that addresses

9. From your experience, which of the following **risk domains** are most underserved by current SRM practices in the battery sector? (Rank 1 = most underserved, 5 = least underserved)

	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
Supply disruption — concentration, geopolitical exposure	☐	☐	☐	☐	☐
ESG incidents — human rights, artisanal mining, environmental	☐	☐	☐	☐	☐
Regulatory compliance — EU Battery Regulation, CSRD, TCFD	☐	☐	☐	☐	☐
Predictive/digital tools — data quality, ML readiness	☐	☐	☐	☐	☐
Financial risk — commodity price volatility, hedging	☐	☐	☐	☐	☐

10. The proposed research focuses only on **Tier 1 cell manufacturers** and **Tier 3 raw material suppliers**, intentionally excluding Tier 2. Does this focus on the two most exposed tiers make sense to you?

- ☐ Yes — Tiers 1 and 3 are the right focus for this research
- ☐ Partially — Tier 2 should also be included because:
- ☐ No — the most important tier for this research is actually

11. In case the answer for above question is **Partially / No**, please provide provide the study / instruments that addresses

12. The proposed research restricts its scope to two risk domains: **supply disruption** and **ESG incidents**. Which, if any, additional risk domains do you believe are impossible to exclude from a meaningful SRM benchmarking exercise?

SECTION C — Benchmark Instrument Design Validation

Purpose: Validate whether the five benchmark dimensions correctly capture SRM maturity in the EV battery sector.

13. The thesis uses five dimensions to assess SRM maturity. Please rate each dimension on how well it captures a real, measurable aspect of SRM in the EV battery supply chain

(1 = Poorly captures SRM reality; 5 = Excellently captures SRM reality)

	Rating -1	Rating 2	Rating 3	Rating 4	Rating 5
D1 — Risk Governance (board oversight, 3 Lines of Defence, risk appetite)	☐	☐	☐	☐	☐
D2 — Supply Disruption Management (concentration risk, BCP, dual sourcing)	☐	☐	☐	☐	☐
D3 — ESG Risk Management (supplier audits, human rights due diligence)	☐	☐	☐	☐	☐
D4 — Predictive & Digital Risk Tools (scenario analysis, ML, dashboards)	☐	☐	☐	☐	☐
D5 — Disclosure & Reporting Quality (TCFD, GRI, sub-tier data disclosure)	☐	☐	☐	☐	☐

14. Are there any **critical SRM dimensions missing / to be added** from this framework that you believe should be included?

Common candidates: geopolitical intelligence, financial hedging, technology obsolescence, regulatory tracking, crisis response speed

15. The thesis uses a **5-level maturity scale** (Initial → Emerging → Defined → Integrated → Optimized). Does this scale reflect the maturity progression you observe in industry practice?

- ☐ Yes — the five levels map well to observable industry behaviour
- ☐ Partially — the following level(s) are unclear in practice
- ☐ No — I would suggest a different scale

16. In case the answer for above question is **No / Partially**, please provide a response below

17. The thesis awards a **Level 4 (Integrated) score** only when a firm provides **"explicit, documented process description"** (Tier A evidence) in a public corporate disclosure. Does this evidence standard appropriately reflect what genuine Level 4 capability looks like — or does it risk systematically underrating firms whose best practices are proprietary and not publicly disclosed?

- ☐ Appropriate — public disclosure evidence is the right standard for a transparent benchmark
- ☐ Too strict — genuinely capable firms will be systematically underrated because
- ☐ Too lenient — public disclosures overstate actual capability because

18. In case the answer for above question is **Too strict / Lenient**, please provide a response below

SECTION D — Case firm score selection Input/ Validation

Purpose: Validate proposed case firms before benchmarking begins and identify blind spots in case selection.

19. The benchmark will be built using **publicly available corporate disclosures** (annual reports, sustainability reports, risk management reports) rather than internal firm data. In your experience, do corporate disclosures reliably represent a firm's **actual** SRM capability?

- ☐ Yes — what firms disclose publicly reflects their internal capability reasonably well
- ☐ No — disclosures consistently **overstate** capability, especially in:
- ☐ No — disclosures consistently **understate** capability, especially in
- ☐ It depends — the disclosure-capability gap is largest at:

20. In case the answer for above question is **No / It depends**, please provide a response below

21. For **Tier 3 raw material suppliers** (miners, trading companies), public disclosures are much thinner. What data sources do you consider the most reliable proxy for assessing Tier 3 SRM capability, in the absence of firm-level disclosures? *Examples: TE/Cobalt Congo reports, GBA Battery Passport pilots, RMI certifications, OECD Due Diligence audit results, journalist investigations*

22. The research proposes to benchmark the following **Tier 1 cell manufacturers** as primary cases. Please indicate your view of each as a representative benchmark case:

	Good representative case	Acceptable but not ideal	Poor choice — reason below
CATL (China — market leader)	☐	☐	☐
LG Energy Solution (South Korea — ESG-intensive)	☐	☐	☐
Panasonic Energy (Japan — Toyota/Tesla dual-OEM partner)	☐	☐	☐

23. If any are options are marked **poor choices**, please explain below:

Also, Are there **Tier 1 cell manufacturers missing** from this list whose SRM practices you consider significantly different from CATL, LGES, and Panasonic — and who should be included? *(Examples: Samsung SDI, BYD, ACC, SVOLT)*

24. CATL Assessment Score v/s Expert Opinion

Initial → 1 | Emerging → 2 | Defined → 3 | Integrated → 4 | Optimized → 5

	Expert Score 1	Expert Score 2	Expert Score 3	Expert Score 4	Expert Score 5
D1 — Risk Governance 4 (Integrated)	☐	☐	☐	☐	☐
D2 — Supply Disruption 3 (Defined)	☐	☐	☐	☐	☐
D3 — ESG Risk 3 (Defined)	☐	☐	☐	☐	☐
D4 — Predictive Tools 3 (Defined)	☐	☐	☐	☐	☐
D5 — Disclosure Quality 4 (Integrated)	☐	☐	☐	☐	☐

25. Provide comments on CATL Scoring (if required)

26. LG Assessment Score v/s Expert Opinion

Initial → 1 | Emerging → 2 | Defined → 3 | Integrated → 4 | Optimized → 5

	Expert Score 1	Expert Score 2	Expert Score 3	Expert Score 4	Expert Score 5
D1 — Risk Governance 4 (Integrated)	☐	☐	☐	☐	☐
D2 — Supply Disruption 3 (Defined)	☐	☐	☐	☐	☐
D3 — ESG Risk 4 (Integrated)	☐	☐	☐	☐	☐
D4 — Predictive Tools 3 (Defined)	☐	☐	☐	☐	☐
D5 — Disclosure Quality 4 (Integrated)	☐	☐	☐	☐	☐

27. Provide comments on LG Scoring (if required)

28. Panasonic Assessment Score v/s Expert Opinion

(Note: Panasonic's lower scores partially reflect disclosure gaps, not necessarily lower capability)

Initial → 1 | Emerging → 2 | Defined → 3 | Integrated → 4 | Optimized → 5

	Expert Score 1	Expert Score 2	Expert Score 3	Expert Score 4	Expert Score 5
D1 — Risk Governance 3 (Defined)	☐	☐	☐	☐	☐
D2 — Supply Disruption 2 (Emerging)	☐	☐	☐	☐	☐
D3 — ESG Risk 4 (Integrated)	☐	☐	☐	☐	☐
D4 — Predictive Tools 2 (Emerging)	☐	☐	☐	☐	☐
D5 — Disclosure Quality 2 (Emerging)	☐	☐	☐	☐	☐

29. Provide comments on Panasonic Scoring (if required)

30. The research cannot benchmark **individual Tier 3 firms** due to data opacity and will instead use a **sector-level composite** (drawing on TE 2024 cobalt reports and GBA Battery Passport pilot findings). Does this approach capture the Tier 3 reality adequately?

- ☐ Yes — sector composite is the only viable approach given data limitations
- ☐ Partially — these named firms/sources would provide better Tier 3 data:
- ☐ No — individual Tier 3 case studies are achievable if you access

31. In case the answer for above question is **No / Partially**, please provide a response below

32. In your experience, are there **exceptions at Tier 3** — major industrial mining companies whose SRM maturity is significantly higher than artisanal and small-scale mining operations — that should be separated from the sector composite?
Example for major industrial mining Companies are - Glencore, SQM, Albemarle, etc.

SECTION E — Simulation Parameter Validation

Purpose: Gather practitioner-grounded estimates for simulation input parameters BEFORE the model is built, reducing reliance on secondary-source approximations.

The research will build a Monte Carlo simulation to model the financial impact of supply disruption and ESG incidents for a representative Tier 1 cell manufacturer. Your input will directly calibrate the model's parameters

33. The thesis models annual critical material procurement spend at **500 Million €** for a representative Tier 1 cell manufacturer.

Is **annual critical (or Tier-3) material procurement spend** plausible as an industry average for Tier 1 cell manufacturer ?

- ☐ Yes — plausible for a Tier 1 manufacturer
- ☐ Too high — a more representative figure would be approximately
- ☐ Too low — a more representative figure would be approximately

34. In case the answer for above question is **Too high / Low**, please provide a response below in **Million €**

35. In your professional experience, what is the approximate probability that a Tier 1 cell manufacturer experiences **at least one significant supply disruption** per critical mineral in any given 12-month period?

Critical Mineral V/s Estimated Annual Disruption Probability

	<5%	5-15%	15-30%	30-50%	>50%
Cobalt (DRC-sourced)	☐	☐	☐	☐	☐
Lithium (Lithium Triangle)	☐	☐	☐	☐	☐
Nickel (Indonesia/Philippines)	☐	☐	☐	☐	☐

36. When a supply disruption occurs, what is a realistic **average duration** before material supply is restored or substituted?

- ☐ Less than 2 weeks
- ☐ 2–6 weeks
- ☐ Option 36–13 weeks (1–3 months)
- ☐ 13–26 weeks (3–6 months)
- ☐ Over 6 months

37. For each week that a critical material supply is disrupted, what percentage of annual material procurement cost does a Tier 1 manufacturer typically absorb as operational loss (production downtime, emergency sourcing premium, contract penalties)?

- ☐ Under 0.5% per week
- ☐ 0.5–1.5% per week
- ☐ 1.5–3% per week
- ☐ 3–5% per week
- ☐ Over 5% per week

38. Based on your experience please provide range for annual material procurement cost

39. In your professional experience, how likely is a **material ESG compliance incident** at Tier 3 (e.g., child labour finding, environmental violation, conflict mineral misreporting) to occur within a 12-month period for a typical Tier 3 cobalt or lithium supplier?

- ☐ Less than 10%
- ☐ 10–25%
- ☐ 25–40%
- ☐ 40–60%
- ☐ Over 60%

40. Please provide an estimate for ESG Probability (Low risk - Base Case - Stress Case)

41. When an ESG incident is discovered and confirmed, what is a realistic **total cost** to the Tier 1 cell manufacturer (regulatory penalty + reputational damage + remediation + sourcing disruption), considering EU Regulation 2023/1542 penalty structures?

- ☐ Under EUR 500K
- ☐ EUR 500K–5M
- ☐ EUR 5–20M
- ☐ EUR 20–50M
- ☐ Over EUR 50M

42. Please provide an estimate for ESG Cost / Penalty ranges (Low - high Case)

43. The simulation will model three specific **supply disruption interventions**. Please rate each on how impactful and realistic it is for Tier 1 firms to implement:

Realistic Impact (1 = Low impact ; 5 = High impact)

	1	2	3	4	5
Dual-sourcing (multiple suppliers per mineral)	☐	☐	☐	☐	☐
Strategic safety stock (buffer inventory)	☐	☐	☐	☐	☐
ESG audit investment (reducing incident probability)	☐	☐	☐	☐	☐
Long-term offtake contracts with governance clauses	☐	☐	☐	☐	☐
Geographic diversification of sourcing	☐	☐	☐	☐	☐

44. The simulation will model three specific **supply disruption interventions**. Please rate each on how impactful and realistic it is for Tier 1 firms to implement:

Realistic Adoption Rate (1 = rarely used; 5 = commonly implemented)

	1	2	3	4	5
Dual-sourcing (multiple suppliers per mineral)	☐	☐	☐	☐	☐
Strategic safety stock (buffer inventory)	☐	☐	☐	☐	☐
ESG audit investment (reducing incident probability)	☐	☐	☐	☐	☐
Long-term offtake contracts with governance clauses	☐	☐	☐	☐	☐
Geographic diversification of sourcing	☐	☐	☐	☐	☐

45. Are there **critical interventions missing** from this list that Tier 1 firms actually use to manage supply disruption or ESG incident risk?

SECTION F — Predictive Tool Adoption Barriers

Purpose: Ground the modular adoption roadmap in real practitioner knowledge of what stops firms from implementing predictive risk tools — before the roadmap is designed.

46. The research will develop a **three-stage modular adoption roadmap** for predictive risk tools:

- **Stage 1** — Structured analytical tools (risk heat maps, scenario matrices) — low cost, low data requirement
- **Stage 2** — Quantitative simulation (Monte Carlo modelling) — medium cost, medium data requirement
- **Stage 3** — Machine learning integration (AI-based risk monitoring) — high cost, high data requirement

Does this three-stage progression match how you observe firms actually adopt risk analytics capabilities?

- ☐ Yes — this sequencing reflects typical capability development paths
- ☐ Partially — firms typically skip Stage 2 and jump directly from Stage 1 to Stage 3
- ☐ The stages are right but the barriers between stages are different from what I'd expect
- ☐ No — the most common progression I observe is (Provide response in below)
- ☐ Other

47. What are the **top three real-world barriers** preventing Tier 1 cell manufacturers from advancing to Stage 2 or Stage 3 predictive risk tools? *(Select up to three)*

Please select 3 options.

- ☐ Data unavailability — internal data not systematically collected
- ☐ Data quality — existing data is inconsistent or unreliable
- ☐ Technical capability gap — no in-house ML / data science function
- ☐ Budget constraints — capital not allocated to risk analytics
- ☐ Organisational resistance — risk function not trusted with predictive outputs
- ☐ Regulatory uncertainty — unclear if tool outputs meet compliance requirements
- ☐ Vendor dependency — lack of open-source / affordable tools
- ☐ Leadership priority — SRM not a board-level strategic priority
- ☐ Other

48. The research proposes these **investment thresholds** for each adoption stage. Do these figures reflect realistic implementation costs for a mid-scale Tier 1 cell manufacturer? (If any figure is off, your suggested range: Stage 1: EUR _ | Stage 2: EUR _ | Stage 3: EUR _

Stage Name	Value (€)
Stage 1 — Structured Analytics	EUR 50,000
Stage 2 — Monte Carlo Simulation	EUR 200,000
Stage 3 — ML Integration	EUR 1,000,000

49. What is a realistic **implementation timeline** for each stage from a standing start (no prior predictive tool capability)? (If any figure is off, your suggested range: Stage 1: EUR _ | Stage 2: EUR _ | Stage 3: EUR _

Stage Name	Proposed Timeline
Stage 1 — Structured Analytics	3–6 months
Stage 2 — Monte Carlo Simulation	12–18 months
Stage 3 — ML Integration	24–48 months

50. For **Tier 3 raw material suppliers**, which single barrier to SRM improvement do you consider most fundamental — the barrier that, if removed, would unlock the most progress?

- ☐ Lack of digital infrastructure (no systems for data recording / reporting)
- ☐ No regulatory mandate requiring disclosure or governance documentation
- ☐ No commercial incentive (certification premium or market access benefit)
- ☐ No accessible capacity building tools (everything is designed for large firms)
- ☐ Jurisdictional governance failures (state-level oversight absent)
- ☐ Other

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